ProbeFlow: Exact Acquisition, Strong Comparators, and the Limits of Useful Conditional Control

Finite theory, strong-comparator development and a posthoc finite-law diagnosis

Abstract

Adaptive acquisition is meaningful only under a declared information, stopping, terminal, cost and comparator contract. ProbeFlow combines exact finite-law certificates with frozen development studies that test their limits. Complete finite routed models exhibit conditional adaptive value, including a reference-relative empirical-law certificate of 2091/205940, while stronger exposed-development static controls win and objective-matched myopic families attain both primary adaptive values. WISDM predictor improvements and terminal calibration do not establish useful acquisition gain. Known-person policy-mixture robustness fails the exposed CAL mean coverage floor.

The new conditional-control study replaces a short catalogue by a full supported paid-token-history class and an equally randomized six-global-order static class. Both share stopping, abstention, free-prior prediction, current-mask terminals, price and coverage floors. A sparse realization-weight LP fits a lower TUNE loss for adaptation, but prespecified participant development CV selects static. On reused CAL both coverage gates pass and adaptive-minus-static loss is +0.018873137736314383. This is no_gain with regression. View alignment uses one TRAIN-only predictor bank; historical exposure remains. A posthoc CAL-label fit with separately checked arithmetic reveals a small finite-class potential advantage, outweighed by larger frozen-policy excess; it is not an independent acquisition gain.

Complete conditions, proofs, participant outcomes, numerical and process limits, failures and source-reading scopes are preserved. Expected cached policies, synthetic arithmetic, oracle diagnostics and posthoc fitted potential are distinct evidence types. No independent-new-person, physical-sensor, reproduced contemporary-method, general POMDP-solver or population-safety claim follows.

1. Introduction

A feature-acquisition system decides whether to pay for another observation, stop with a prediction, or abstain. The answer depends on what the current history reveals, how unsuccessful attempts are charged, which terminal decisions remain available, and what the comparator is allowed to do. A fixed-order policy can still stop after an observation, choose a history-dependent prediction and randomize before an episode. Removing these rights can create an apparent benefit of adaptation that disappears under a stronger control.

We study the additional right to choose the next observation from the paid history. The theoretical starting point is deliberately finite: declared latent laws, a complete observable policy grammar, shared terminal and budget rights, and exact rational evaluation. The finite setting permits complete witnesses and decisive counterexamples. It also makes the validity boundary explicit: an exact optimizer does not show that a target population belongs to the supplied law set.

The empirical question is harder. Prediction improvement, useful prediction coverage and acquisition improvement are distinct. A predictor can correct many errors but remain too inaccurate for its paid cost. A selector can reduce accepted-error mass by abstaining on most cases. A constrained mixture can satisfy every fitted coverage condition and fail the same mean condition on exposed development people. We therefore report error, coverage, purchases, correction, harm and feasibility separately.

This article unifies the complete finite-law theory, the MHEALTH development audit and the WISDM predictor, terminal-combination, observable-acquisition and coverage-transport studies. The scientific contribution is a fully specified comparison and evidence chain: same-world reference contrasts; exhaustive finite policy constructions; explicit coverage convexification; strong randomized static baselines; and retained failures that delimit applicability. Standard dynamic programming, robust policy improvement, mixture learning and linear programming provide its foundations; no general novelty of those principles is claimed.

The latest question is conditional rather than catalogue-limited: can optimized observation-dependent next-axis choice beat a fully matched randomized global-order controller? We connect the richer fitted class to the earlier finite theory, then test selection and transfer separately. A better training optimum is not a development benefit, and a development ranking is not a new-person result. The article presents this as one argument rather than a sequence of replacement abstracts.

The technical supplement is organized by mathematical and experimental function. Its complete proofs and counterexamples are in Supplement S1–Supplement S5; MHEALTH methods and full outcomes in

Supplement S6; WISDM construction and detailed participant results in Supplement S7–Supplement S12; and reproducibility and source-reading limits in Supplement S13–Supplement S14. The article and supplement are an editorial reconstruction of frozen evidence, with no new scientific execution.

2. Related work and the comparison contract

Learned active feature acquisition includes variable-size set encoding [1], generative surrogate models [2], stochastic encodings [10], Learning-To-Measure [7], and non-myopic pathwise policy gradients [11]. Preserved readings range from official abstracts to selected versioned sections. They motivate method categories; none is a reproduced empirical comparator here. Set-representation order invariance does not imply that sequential action order is irrelevant under stopping, cost, availability or failure.

Retrospective acquisition evaluation additionally depends on label availability, missingness and positivity. The preserved static- and time-varying-feature readings [9] [8] distinguish these identification assumptions from acquisition optimization. A static feature setting can contain adaptive policies. Declaring a finite uncertainty set does not identify it from masked observations, and an exact certificate does not supply unverified positivity or a missing-not-at-random identification argument.

Reference-relative comparison has established robust baseline-regret and policy-improvement precedents [14] [12]. Baseline bootstrapping and distributionally robust policy learning also precede the conditional fallback constructions [15] [17]. Our small same-world matrix proofs and finite-portfolio diagnostics do not inherit their finite-sample, confounding or implementation guarantees. The unlabeled counterexample is an explicit instance of the established domain-adaptation identification limit [18].

Finite-horizon piecewise-linear planning is foundational [13]. Constrained control, performance-vector convexification, LP duality and randomization are classical [74]; constrained partially observed planning is also prior art [75]. Those book and article references were inspected only within the recorded metadata/abstract scopes, not as full theorem audits. We prove the finite statements used here directly. Nonrectangular ambiguity is not silently replaced by independent conditional adversaries; the robust-DP source scope is explicitly limited [71].

Paid terminal combination is related to Super Learner, conditional ensemble selection, selective prediction and probability calibration [58] [59] [60] [61] [62]. These references have different inspected editions and depths. For example, the Super Learner record supports its abstract-level cross-validation/oracle description, not an uninspected specific weighting proof. Our paid-submask decomposition identifies the distinction between selecting an expert label and producing a new fused label; it does not establish a new general ensemble theorem.

Repeated selection on development data can bias assessment [3]. That concern is observed here rather than hypothetical: MHEALTH C/DEV and WISDM TUNE/CAL are exposed, and global FINAL independence is unresolved. Bibliography entries preserve source-specific reading scopes in Supplement S14 and REFERENCES_SCOPE.json. Related work is bounded comparison, not a comprehensive novelty certification or a claim about unread passages.

The conditional-control representation also belongs to established realization/sequence-form and constrained partially observed planning traditions. The source records for SciPy/HiGHS [81] [82], the selected CPOMDP passages [83] and the metadata/abstract-only sequence-form record [84] retain their exact inspection scopes. We construct and test this small finite instance directly, without claiming a new LP principle or clearance of general novelty.

3. Information, notation and evidence units

A latent world contains a label Y and all potential query responses. A public history h records only observations already paid for, prior actions and the remaining budget. At history h a policy may make an allowed prediction, abstain, or attempt a legal unbought query. Offline fitting and planning can use the declared training worlds. The online action function cannot use hidden labels, future responses, participant identity or an unobserved world index unless the model explicitly grants that observation.

We write A, F and M for adaptive query-choice, fixed-order and a precisely specified myopic family. Fixed subsets are a separate class. A fixed order is selected before observations but may stop at any allowed history; it cannot choose a different order after seeing a root token. A forced subset continues in its declared canonical order until completion or a shared support fallback. An independent ex ante coin chooses a complete policy, not a new history-dependent gate.

Throughout this article C is coverage, E is wrong-prediction probability per all cases, K is expected paid acquisitions and J is total loss. Selective error is E/C only when C is positive. The MHEALTH source also used A for an abstention count and C for a calibration split or quote count; in this article those quantities

are written explicitly in words or scoped to their source tables. Lowercase b is a fixed reference loss vector. Γ denotes guaranteed same-world improvement over that reference. The symbols θ, α and ρ always retain their locally stated law-mixture, design or uncertainty roles.

For people with different numbers of windows, rates are formed within each person and then averaged equally. Pooled-window rates are reported separately. Repeated policy evaluations, overlapping windows and repeated predictions are not independent people. Rational finite laws, mathematical synthetic checks, cached-data experiments, independent saved-output audits and external author reports remain separate evidence types.

Figure 1 shows the operational information boundary. It is a cooperative program interface tested in frozen code and records, not an OS isolation or security certificate. In the binary theory the normal model has no mandatory first query. In MHEALTH the initial token is paid. In the anchored WISDM primary comparison Y is bought first. The ledgers therefore share a loss form without having interchangeable denominators or initial conditions.

In the new conditional-controller study an episode starts before any paid axis. The fixed TRAIN prior can predict freely; otherwise each purchased cumulative-mask token contains argmax class and a confidence bin. Only visited masks are visible, not every possible paid submask. This differs from the calibrated-.45/Y-first predecessor, so cross-study changes are not an isolated causal acquisition effect. Within the new study both classes receive identical rights.

$$J_P(\pi) = E_P(\pi) + \left(\frac{2}{5}\right)(1 - C_P(\pi)) + \left(\frac{1}{20}\right)K_P(\pi)$$

Common logical-unit loss in the empirical studies: purchase cost 0.05, wrong prediction cost 1 and abstention cost 0.4. The finite routed model replaces 1/20 by its nonnegative cost parameter λ.

$$\overline{J}(\pi) = \left(\frac{1}{G}\right)\sum_{g=1}^{G} J_g(\pi), \quad \frac{E}{C} = \text{undefined} \quad \text{when } C = 0$$

Equal-person summaries are primary. The ratio of macro masses is distinct from the mean of person-specific selective-error ratios.

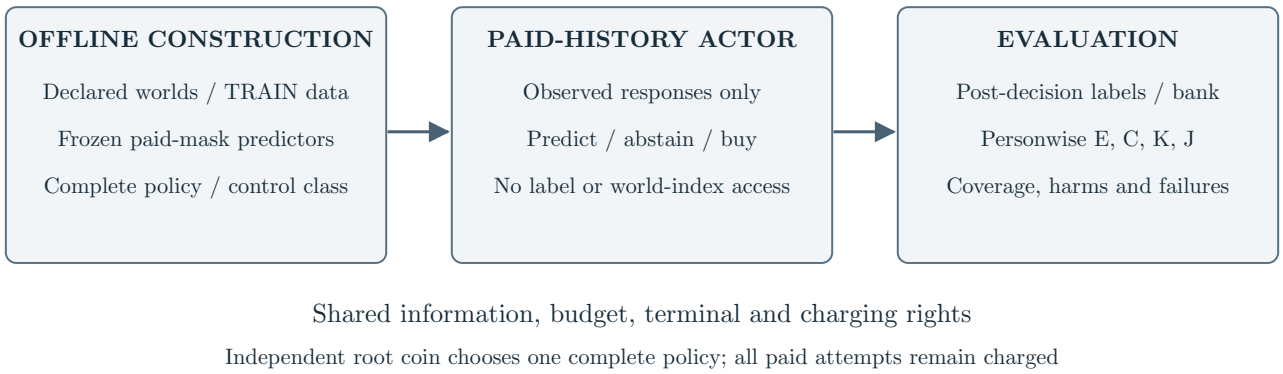


Figure 1: One information contract from offline law/model construction to paid-history execution and post-decision evaluation. The common loss decomposes error, abstention and paid attempts; controls receive matching information and terminal rights.

4. Unified construction and verification procedure

The procedure has four distinct objects: an input law or cached predictor bank; a legal observable policy class; an objective and comparator selected under declared information; and a frozen evaluation rule. Exact finite optimization and learned risk estimation are different backends to this contract. A backend returns an action table or mixture together with its support, objective components and failures. An unknown or exhausted computation is not a zero-gain result.

In a complete finite grammar, a terminal is a permitted prediction or abstention; a query node contains one legal subtree per possible response. Attempts reduce the remaining budget and remain charged when responses fail. All legal fixed orders and all allowed terminal choices are retained. Numerical deduplication or dominance reduction can accelerate an objective, but raw rows and original tree indices remain available for all-row verification. The routed binary normal grammar has 1,326 adaptive trees and a structural union of 840 fixed trees; singleton erased responses give 30 in each class.

In cached empirical planning, support is an explicit restriction on acquisitions. MHEALTH requires at least 30 compatible T rows and both fitting people in the current cell and every positive successor. The later observable WISDM construction requires at least 16 rows and two training people. A sparse or unseen history uses the frozen terminal fallback; it does not silently drop history components, read an unbought feature or fabricate a smoothed continuation. Different support rules define different policy classes.

A coherent empirical world gives every row its original fixed person-normalized weight. Histories partition these rows, and conditional risks and query transitions are derived from the same masses. Rebalancing people to equal weight after each observation changes the joint law. Combining separately worst conditional risks and occupancies can likewise create a fictitious law. The joint-world constructions retain unconditional world-specific performance vectors through branches before taking a robust maximum.

For learned continuations, whole-pipeline participant exclusion rebuilds terminal-risk models, continuation models, targets, weights and caches without the held-out participant. This closes the specified computational label path. It does not remove optimism from fitted minima among the remaining people. Fully recursively excluded targets are a distinct, more expensive construction. Supplement S8 gives both dependency conditions, the exact counterexample and matched protocol.

The same algorithmic contract applies to explicit failure. An invalid source hash, unsupported state, violated convergence criterion, incomplete process observation or infeasible LP remains visible with its denominator. This manuscript runs no learner or scientific verifier again; it reconstructs their frozen results and retains the original validation scope.

Algorithm 1. Construct, freeze and compare a paid-history policy
Require: Declared worlds or hash-bound cached predictors; legal paid observations, terminal actions, support and budget rules. Fixed loss, selection split, strong comparator class, coverage constraints and explicit resource/failure contract.
Ensure: Frozen complete-policy table or common ex ante mixture with exact support and per-world/person E,C,K,J. A scoped certificate, no_gain, infeasible, failed or unknown result; preserved provenance and denominators.

1. Validate source identity and the complete information/charging contract; separate actor-visible observations from evaluator truth.
2. Enumerate legal finite policies, or fit terminal and continuation estimates using only the permitted participants and paid features.
3. Derive risks and transitions from one coherent law when that is the declared construction; preserve support fallbacks and failed attempts.
4. Optimize candidate and strongest matching controls under identical objectives, stopping rights, randomization and coverage constraints.
5. Validate feasibility and exact or numerical certificates over all original rows; freeze the reference vector, policy IDs, weights and tie rule.
6. Evaluate the frozen choices under the named law or held-out development partition; do not retune after failure.
7. Report all error, coverage, acquisition, paired transition and failure components, including ties and infeasible comparisons.
8. Return an explicit incomplete result when required proof, process, data or resource evidence is unavailable.

5. Same-law finite certificates and an exact acquisition envelope

The basic finite model has independent fair bits B0 and B1, route C, and target $Y = B_C$. Endpoint P0 fixes C=0 and endpoint P1 fixes C=1; within either endpoint the four bit pairs are uniform. Query 0 reveals C, query 1 reveals B0 and query 2 reveals B1. The actor sees neither the endpoint index nor an unbought coordinate. With at most two distinct attempts, it shares prediction, abstention and independent randomization rights with every fixed-order comparator.

For a finite two-column loss matrix, the absolute robust value minimizes the largest endpoint loss. Reference-relative improvement instead fixes one selected reference vector b first and minimizes the worst candidate-minus-b contrast at the same endpoint. These quantities cannot generally be obtained by subtracting separately optimized worst losses. For example, reference losses (0,1) and an added row (1/2,1/2) reduce absolute minimax loss from 1 to 1/2, yet every positive use of the added row harms endpoint 0; uniform same-reference gain remains zero.

An extreme optimum of a nonempty two-column epigraph LP has positive support on at most two rows. The solver can therefore check every pure row and valid equalizing two-row crossing, while the dual checks endpoint weights and line intersections. Exact primal feasibility, every original-row dual inequality and equal

rational values give a certificate. The small-support existence statement is specific to two law columns without extra side constraints; it does not justify a two-row solver for later coverage LPs.

The routed model admits an all-cost proof. Under the symmetric route law, a no-observation prediction has error one half and abstention costs 2/5. One observed bit gives posterior error 1/4. A symmetric mixture of C-first fixed orders predicts perfectly on its matching route and abstains on the other, using 3/2 expected queries. Adaptive routing reads C and then the relevant bit, using two queries with no error. Comparing the first-query and terminal cases gives the exact envelopes below.

The gain is positive below λ=1/5 and zero thereafter. All nine designed cases completed: four routed gains (1/5, 7/40, 1/8 and 1/15 at costs 0, 1/20, 1/8 and 1/6) and five no_gain cases (boundary cost, expensive cost, erased responses, a static target and known route). The randomized static reference is stronger than a pure order: at costs 1/20 and 1/8 it removes 1/40 of the apparent deterministic-baseline advantage. These are selected exact conditions, not an empirical success rate.

The full grammar, endpoint tallies, ten fixed-order lower bounds, degeneracies, dual witnesses, every designed case and the separate interval-width lineage are in Supplement S1 and Supplement S2. The interval-width objective has its own complementarity obstruction; reducing a conditional interval or computing it faster is not the same outcome as lowering prediction loss.

$$V_K = \min_{w \in \Delta(\Pi_K)} \max_{v \in \{0,1\}} \sum_i w_i L_{iv}, \quad K \in \{F, A\}$$

Both classes permit independent ex ante mixtures of complete legal policies.

$$\Gamma_b = \max_{w \in \Delta(\Pi_A)} \min_{v \in \{0,1\}} \left(b_v - \sum_i w_i L_{iv} \right)$$

Candidate and reference are compared under the same endpoint. Reference inclusion implies the guaranteed gain is nonnegative; zero means retain the reference.

$$V_F = \min\left(\frac{2}{5}, \frac{1}{4} + \lambda, \frac{1}{5} + 3\frac{\lambda}{2}\right), \quad V_A = \min\left(\frac{2}{5}, 2\lambda\right)$$

Exact nonnegative-cost envelopes under the declared routed-bit assumptions. Fixed breakpoints are 1/10 and 3/20; adaptive abstention starts at 1/5.

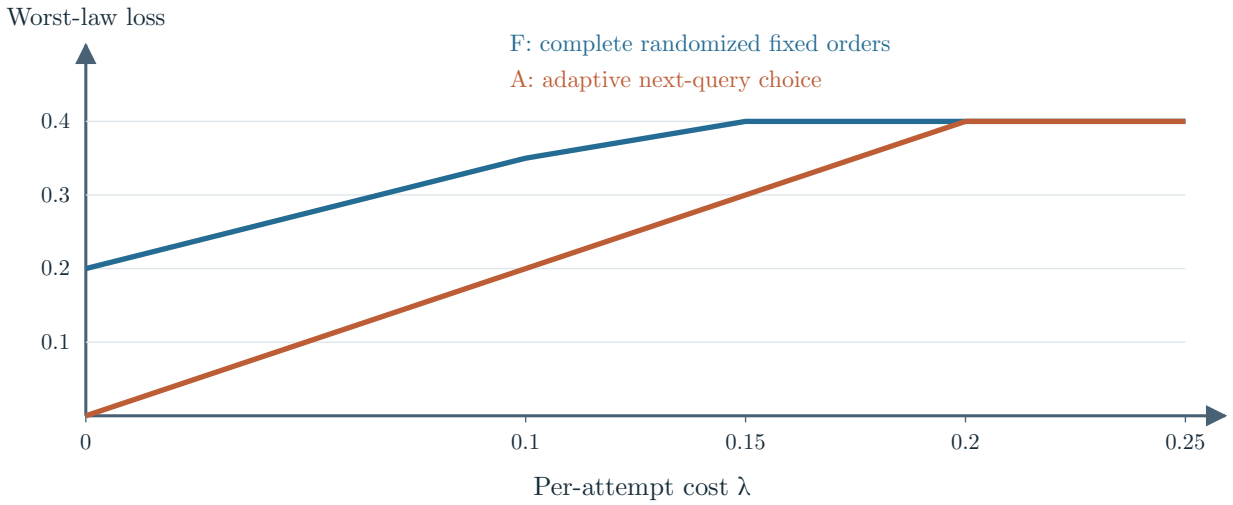


Figure 2: Exact conditional routed-model envelopes from the frozen proof, in logical loss units. The fixed class includes randomization and history-dependent stopping. The gap vanishes at cost 1/5; this curve is not a measured sensor-cost trend.

6. Strong fixed comparators, acceptance and paid fusion

A selected reference and a law-specific best-fixed envelope pose different quantifiers. If uncertainty leaves the target mapping ambiguous, a common policy may fail to dominate the best fixed action selected separately in each law even when it minimizes robust absolute loss. The paid-perfect-observation construction and its if-and-only-if boundary in Supplement S3 quantify this obstruction, including every mixture. Simultaneous estimation bounds require their stated uncertainty event; neither baseline selection nor an empirical optimizer supplies it automatically.

Coverage constraints change the feasible comparison. A fixed-paid predictor with accuracy a and randomized acceptance coverage c has sharp accepted-error lower bound max(0,c-a). With person-specific accuracy bounds and person coverage floors, the sharp envelope is formed person by person before weighting. This is a correctness-event oracle: it can accept precisely the cases on which a predictor is right and is therefore not a deployable confidence selector. Deterministic finite samples also have a coverage lattice. The corresponding cost exclusion can rule out a specified fixed-paid family without ruling out a sequential adaptive alternative.

Let U denote the event that at least one eligible paid expert argmax is correct. An expert-label selector cannot be correct outside U. A general probability mixture or class-bias calibration can output a class that no raw expert predicts. It can therefore exceed the raw argmax union. The exact accepted-error identity separates accepted union mass V, selector failure inside the union F and fused correctness outside it G. Positive G alone is not evidence of multi-expert value: an own-mask calibration can also produce it.

The paid-fusion theory gives explicit calibrated counterexamples to choosing the largest confidence and states the recognizability, sufficiency and transport premises needed to bound the recoverable selection gap. Those premises are unverified for the empirical WISDM bank. Even individually calibrated experts need not make maximum confidence optimal after history-dependent selection. The four-cell bounds, sharpness conditions, eight-atom adaptive witness, fusion counterexample, decomposition proof and complete comparison table are in Supplement S4.

These results motivate a strong empirical terminal comparison: hold purchased information and acceptance rights fixed, compare calibrated own-mask, arithmetic/log pooling, maximum confidence and learned convex weights, and separately measure coverage. A label-informed correctness union diagnoses a ceiling for a stated selector class; it is not an achieved acquisition gain or a performance forecast.

$$E \geq \max(0, C - a), \quad J \geq \max(0, C - a) + \left(\frac{2}{5}\right)(1 - C) + \left(\frac{1}{20}\right)K$$

Sharp randomized correctness-selection lower envelope for a fixed paid action; a is its unconditional accuracy and K its fixed purchase count. Valid accuracy bounds and deployment-recognizable acceptance are external requirements.

$$E = C - V + F - G$$

Accepted-error identity: V is accepted expert-union mass, F accepted terminal error inside that union and G accepted terminal correctness outside it. The equality holds before person averaging.

7. Joint worlds, common randomization and constrained certificates

An uncertainty set consists of complete joint worlds. For each observable deterministic policy and world, preserve the complete E,C,K,J vector. One world must govern both candidate and comparator and every branch of an episode. A hidden world index is not available to the actor. Optimizing one occupancy distribution independently in each world can violate this nonanticipativity requirement.

An independent root coin is retained through the whole episode. Mixing the complete-policy vectors gives expected performance exactly. Coverage is imposed after convexification: discarding every individually infeasible pure policy can remove a feasible mixture. If no mixture satisfies the linear constraints, randomization cannot rescue the stated library. The feasible set is common across worlds and groups, and its constraints are expected rates rather than sample-path quotas.

The strongest static benchmark here has equal randomization and stopping rights. For each world, minimize static loss over the same jointly coverage-feasible mixture polytope to obtain its envelope b_v. Then minimize the common candidate's worst same-world excess. A negative optimum certifies strict improvement in the declared set. A positive minimax excess is still a failure to dominate that envelope, even if it is the smallest available excess.

In the observed-mode default example, adaptive J=1/10 and randomized static J=13/50, so the gap is 4/25. A weaker deterministic static control gives 31/100 and overstates the gap as 21/100. At group floors 3/4 the randomized static optimum is 57/200; at full coverage it is 31/100. Under paired noiseless/noisy worlds the guaranteed adaptive gain shrinks from the deterministic comparison's 6/100 to 1/100. The same adaptive policy equals all-abstain loss in the noisy world, so strict improvement over all-abstain is false there.

The ten exact constrained cases also retain no-information, hidden-group, hidden-mode, high-cost, XOR and enlarged-ambiguity negatives. Adding an inverted world preserves observable coverage yet removes adaptive gain; both adaptive and common-static excess optima become 7/100. A frozen nominal policy evaluated without retuning in an excluded world has loss 3/5, worse than all-abstain by 1/5. These examples delimit declared-world inclusion, not a population failure probability.

Two counterexamples explain why local shortcuts are insufficient. With no observations and opposite constant labels in two worlds, separate world-specific policies report zero loss; the best common coin has worst loss 1/2. With observed histories, a global coverage floor can be met by accepting only an informative history; replacing it with the same floor at each history is unnecessarily restrictive. Complete proofs, all 43 LP outcomes and exact all-column certificates are retained in Supplement S3.

$$\mathcal{W} = \left\{ w \geq 0 : \sum_i w_i = 1, \quad \sum_i w_i C_{vgi} \geq c_{vg} \quad \text{for every } (v, g) \right\}$$

One common mixture polytope, formed before optimizing and after retaining all legal pure components.

$$b_v = \min_{u \in \mathcal{W}_F} \sum_i u_i J_{vi}, \quad R_A = \min_{w \in \mathcal{W}_A} \max_v \left(\sum_i w_i J_{vi} - b_v \right)$$

The equally randomized static envelope and same-world common adaptive excess. Additional coverage constraints require the general epigraph LP, not the two-row theorem.

$$J_v(w) = \sum_i w_i J_{vi}, \quad C_{vg}(w) = \sum_i w_i C_{vgi}$$

Linearity follows from one independent coin over complete observable policies. It does not require independent component outcomes on a case.

8. Conditional transport and structure-aware fallback

The MHEALTH empirical-law certificate is conditional on its two training-group laws. Fix the reference and an outer mixture. If at one common nominal mixture parameter the candidate contrast is at most minus Γ and every positive-weight component's target-minus-nominal contrast residual is bounded by δ, linearity gives excess at most -Γ+δ. Separate absolute risk residuals bounded by η imply the looser -Γ+2eta. The constants are sharp in the stated bounded constructions; component outcomes need not be independent.

An aggregate portfolio projection is a retrospective diagnostic of named risk vectors. It is not a prospective confidence radius and does not validate unavailable deterministic-component residuals. In the exposed MHEALTH audit, C and DEV units overlap and reuse people. The nominal paired policy worsens participant 7 while improving participant 8; a macro gain is not person safety. The complete exact projection intervals and all nine overlapping units are in Supplement S5.

Unlabeled data cannot resolve every gain sign. A frozen predictor returns 0 on one constant initial observation and 1 after one constant acquired observation. At cost 1/20, labels constantly equal to 1 give candidate-minus-baseline -19/20, while labels constantly equal to 0 give +21/20. Every unlabeled history is identical in distribution. This toy illustrates identification failure; it is not a claim that either law is a member of the actual MHEALTH grammar.

Under a separately assumed joint-law total-variation radius, policy structure can tighten an additive bound. The frozen eleven-name portfolio yields $B(\rho) = \min\big(0, -\frac{2091}{205940} + \big(\frac{5989}{2942}\big)\rho\big)$, compared with the fair uniform nonbaseline sensitivity 11/5. The structure-aware zero-boundary radius is 2091/419230 rather than 2091/453068, an exact relative widening 2417/29945 (about 8.07%). The selected proposal is unchanged before fallback. This is improvement of a conditional upper certificate, not a prediction win or a sharp unrestricted DRO optimum.

The larger-world proof permits arbitrary per-sample predictions across distinct terminal masks while retaining identical-mask consistency; the same sensitivity bounds remain valid. Sharpness is confined to the original finite-token construction. No target TV radius is independently justified, calibrated from unlabeled observations or estimated from C/DEV. Retaining the baseline when a valid target bound fails is a conditional use rule, not an implemented population-safety guarantee.

$$J_Q(\pi) - J_Q(b) \leq -\Gamma + \delta \leq -\Gamma + 2\eta$$

The final inequality uses a contrast-residual bound at most twice the separate absolute-risk bound. Each statement requires the common-reference, common-loss and common-nominal-law premises proved in the supplement.

$$\mathrm{TV}(Q^\star, P_\theta^\star) \leq \rho, \quad B_s(\rho) = \min_{w \in \Delta_{11}} \max_{g \in \{5,6\}} \sum_i w_i \big(d_{ig} + \rho s_i\big)$$

Joint space includes observations, label and per-mask predictions. The target-radius assumption is external; each sensitivity bounds its selector contrast oscillation.

9. Data, window construction and frozen predictor banks

MHEALTH and WISDM serve different purposes and remain separate. The MHEALTH diagnostic reuses a historical token/prediction bank. Fitting people 5 and 6 have 514 complete rows, split by the preserved index-block rule into T=322 and C=192. Exposed development people 7 and 8 have 266 and 258 windows. The lineage has used all ten underlying people; there are zero independent-final people. It is a cached mechanism audit, not a new raw-sensor benchmark [4].

The MHEALTH bank permits masks 1,3,5,9,7,11,13; mask 15 is outside its domain. The initial token and every additional quote cost 1/20. There are 38 eligible token states per estimator. Empirical terminal risk uses compatible T error, with 1/2 for an empty set. The calibrated variant blends compatible C errors with 64 times empirical risk, while retaining T transitions. That statewise calibrated risk and kernel are not asserted to be conditionals of one law. The separate coherent-law diagnostic is not retroactively substituted for this variant.

The public WISDM source identifies 51 people, 18 activities and phone/watch accelerometer/gyroscope blocks. Original timestamp inspection did not establish a common four-stream clock. The accepted same-row alternative uses only x,y,z of the phone accelerometer as three logical information units. They are not independently priced physical sensors. The official metadata and existing local description/README scopes are preserved [28] [29] [30].

The retained protocol constructs activity-homogeneous windows of 100 ordered samples under fixed gap, reversal, nonfinite and short-segment rejection rules. One hundred samples are not a uniform physical duration. Activity labels define offline segments, so online boundary detection is not evaluated. TRAIN has 25 people/24,880 windows; TUNE 8/7,845; CAL 8/6,437. These 41 people and all three partitions are development evidence. Ancillary FINAL eligibility exposure and an incomplete historical JSON-path exclusion leave global independence unresolved; this manuscript does not open FINAL payloads.

The original representation has six descriptors per axis: mean, population standard deviation, minimum, maximum, RMS and mean absolute successive difference. The fixed extension adds autocorrelation, spectral-shape and centered sign-change descriptors, giving 54 features. It preserves the original 18-column layout before the appended 36 columns. Paid-axis selection must take the sorted union of its original and appended columns; contiguous slicing would leak unpaid axes. FFT bins describe ordered samples, not hertz, cadence or physical energy.

Saved ExtraTrees models use 128 trees, maximum depth 12, minimum leaf size 5, max_features=1, no bootstrap, seed 6101 and one thread. The full-mask 54-feature classifier was fitted once on TRAIN; the later shared family adds six proper-mask TRAIN fits. HGB is a separate seven-mask intervention with 100 multiclass boosting iterations and 18 classes. Equal model settings do not imply equal realized capacity, and HGB tree counts do not provide a matched-capacity comparison. Supplement S7 preserves source custody, all window denominators, exact formulas, unpaid-feature checks and all participant results.

For conditional control, the same saved TRAIN-only ET54 mask predictors and TRAIN prior produce both TUNE and CAL views. No predictor, calibrator or normalizer is fitted again. This removes the older OOF-versus-refit view mismatch from the new comparison only; it does not restore untouched data after repeated development, representation selection or unresolved FINAL exposure.

Table 1: Distinct evidence units and split sizes. Rows must not be pooled into one performance table.

Setting	Fitting / selection	Reported evaluation	Unit and exposure
Routed finite laws	Two declared endpoints; no fitted samples	Nine designed conditions	Exact mathematical worlds
MHEALTH cache	T: 322 rows; C: 192 rows (people 5,6)	DEV: 524 windows (people 7,8)	Two exposed DEV people; independent final 0
WISDM windows	TRAIN: 25 / 24,880; TUNE: 8 / 7,845	CAL: 8 people / 6,437 windows	Exposed development; FINAL independence unresolved
Constrained theory	Ten declared finite scenarios	43 LPs; one excluded-world diagnostic	Exact conditions, not sampled people

10. Matched empirical methods and selection rules

Each empirical comparison freezes its shared predictor bank, terminal rules, allowed observations, budget, support and strong controls. A useful WISDM choice requires mean person coverage at least .50 and every person at least .25. Beating free all-abstain J=.4 is a separate cost-performance condition: all-abstain violates useful coverage. A candidate and reference must both pass the relevant realized gate before a useful-domain loss comparison is called a gain.

The participant-excluded learned acquisition study compares purchased raw features with purchased raw features plus cached confidence/margin summaries. It never observes confidence from an unpurchased mask. Thirty-six builders fit seven terminal and 21 query regressors each, totaling 1,008 risk fits. Each outer TUNE assessment rebuilds the pipeline on the other seven people; the final CAL learner uses all eight TUNE people. The former is not an outer assessment of the exact latter learner.

The HGB fixed-mask study uses the same paid-submask terminal rights as its frozen ET comparator. A terminal may select among already paid submask predictions, including the free prior, with fixed tie rules. Thresholds are 0,.05,...,1 plus explicit all-abstain. TUNE chooses using unrounded J, then K, then -C, then exact ID. Those choices are applied unchanged to exposed CAL; the arrays can be loaded before selection, so the record does not claim access-based CAL blinding.

The paid-fusion attempt planned 55 participant-crossfit/refit multinomial learner calls but stopped at the first nonconvergent fit. It is a completed failure report, not an efficacy trial. The later bounded alternative fits seven 18-class bias calibrators and four convex paid-submask mixtures in each of five scopes: four held-out TUNE participant-pair folds and one all-TUNE refit. Thus 35 bias and 20 simplex calls complete. Their class-bias map may change argmax, unlike scalar temperature scaling. Positive smoothing, a fixed regularizer, gradient residuals and a simplex supporting-plane gap define numerical acceptance; solver success alone is insufficient.

Observable acquisition freezes the calibrated confidence-.45 own-mask terminal. At paid mask M it sees only the argmax class and confidence bin of each nonempty paid submask; bins use cuts .45,.60,.80. It does not see raw sensor values, full unbought vectors, labels or person/row identity. A coherent TUNE row world supplies both conditional errors and transitions. Full-depth, limited-depth and receding-myopic policies share the same support and terminal. The primary budget 3 includes mandatory Y, so anchored depths 2 and 3 must agree. Six global orders, fixed subsets, fixedY45 and a uniform-order coin remain explicit controls.

The nominal coverage optimizer uses an eleven-ID static library and a fourteen-ID enriched library adding three frozen depth policies. IDs that become aliases after Y-first normalization remain in the input and tie rule, not as independent algorithms. One common coin is optimized over the complete library under the same fitted mean/person floors. The later transport optimizer changes only the known-person uncertainty set and matched objective. It does not create a new observation-dependent gate, retrain a predictor or infer a new person's law.

Figure 3 shows the dependencies that must remain fixed to attribute an acquisition effect. Representation, terminal calibration, sequential routing and common-policy randomization are separate interventions. Their performance values cannot be merged across changed predictors or used to turn a later negative comparison into a positive acquisition result.

The new comparison uses the full supported history-action class rather than the earlier eleven/fourteen-ID policy catalogue. Four participant-CV fits and four final paired fits use three lexicographic LP stages each. Fitting histories, coefficients and support exclude the held-out fold; the locked selector chooses static on OOF development. Both final classes are then fitted to all TUNE people and locked before CAL. Details and every primary participant are in Supplement S15.

Algorithm 2. Freeze a useful-coverage comparison on development people
Require: Shared paid-mask predictions or observable-history tables, TRAIN/TUNE/CAL identities and fixed person denominators. Candidate/control catalogues, mean floor .50, person floor .25, common loss and a prespecified tie rule.
Ensure: TUNE-locked candidate/control IDs or weights and complete CAL expectations. Separate candidate feasibility, reference feasibility, delta J and free-abstention comparison; no invented outcome for an infeasible problem.

1 Validate paid-feature and paid-submask rights; keep labels and person IDs out of deployed actions.
2 Build held-out TUNE predictions using the stated whole-pipeline or participant-pair exclusion; keep all-TUNE refit outputs distinct.
3 Retain all legal deterministic components, including those individually coverage-infeasible when common mixtures are permitted.
4 Optimize both candidate and strongest matched static controls using TUNE only; apply the identical objective, coverage constraints and tie rule.
5 Lock exact IDs, weights, numerical certificates and aliases before constructing CAL metrics.
6 Compute per-person E,C,K,J and pooled counts without changing the frozen selection; charge acquisitions on abstained cases.
7 Check the original CAL mean and person floors with unrounded values before a useful-domain comparison; label infeasible or reference-infeasible cases explicitly.

8 Preserve all budgets, controls, paired harms, failed attempts, unobserved process tails and unresolved FINAL independence.

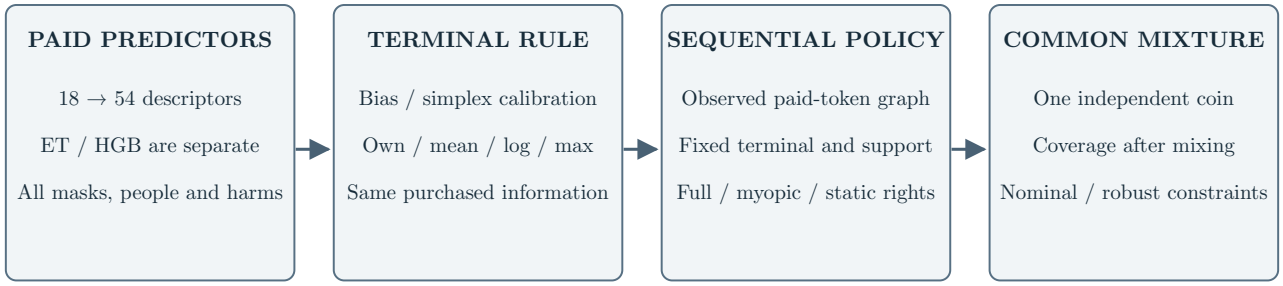


Figure 3: Distinct intervention layers in the empirical comparison. Each later policy uses a frozen shared dependency and matching controls. A predictor gain, a calibrated terminal gain and an acquisition-policy gain answer different questions; development exposure persists through every layer.

11. Conditional full-history control and the matching global-order class

Each original row is one coherent world containing fixed potential cached summaries for all masks. Offline fitting may use its label and person group to construct equal-person loss and coverage coefficients. The deployed actor receives neither label, identity, row index nor unbought output. It sees the order of bought axes and every past current-cumulative-mask token. A token is cached argmax class and one of four confidence bins, with equality at .45/.60/.80 entering the higher bin.

Root variables choose free prior prediction, free abstention or a first axis. Subsequent prediction/buy variables are products of policy action probabilities only. They exclude environmental observation probability. Compatible-row frequencies from the same original law enter the linear C/E/K coefficients. Rebalancing people after each observation would change that law and is not done.

For positive parent mass each child-to-parent ratio is a legal behavioral probability and the leftover is abstention. Zero parent implies all children zero and that history is unreachable. Induction on the complete paid history proves equality between row-wise actor expectations and the LP coefficients. Only TUNE-supported histories are stored; the prespecified shared fallback immediately predicts current-mask argmax on an unsupported CAL history and buys nothing further.

The optimized class is deliberately limited to these current-mask tokens. It does not expose raw paid features or all possible already-paid submask outputs. The mathematical optimum cannot be extended to every available raw feature, richer posterior or omitted terminal expert. Numerical certificates are checked to1e-7; the real-data optimum is not asserted as an exact rational theorem.

Static chooses one of all six complete orders before any observations. It can predict, stop or abstain after every token, and can randomize at every legal decision. For order o=(A,B,C), lambda_o is its ex ante mass. At first-history s the continued mass u_o(s) is at most lambda_o. Root-axis mass is the sum of the two orders beginning with A.

Let r_o(s)=lambda_o-u_o(s). Flow feasibility gives p(s)<=sum r_o(s). Allocate prediction mass p_o(s)=p(s)r_o(s)/sum r, or zero when sum r is zero. Conditional on the preselected order, continue with u_o/lambda_o, predict with p_o/lambda_o and abstain with the remainder. After B the order identifies the sole remaining C, so subsequent ratios are unchanged. This implements every coupled LP solution with an order selected before observing s.

Conversely any such fixed-order actor yields feasible coupled realization weights. The static LP is therefore the full matching fixed-order class on this support, not a forced-order or fixed-threshold surrogate. Adaptive contains it, so the TUNE adaptive optimum cannot exceed static. The small independent Fraction oracle gives J=1/10 versus1/8 in a synthetic routing example; those values validate the representation and are not empirical performance.

The full static class contains all six orders even when the fitted optimum gives mass one to Y-X-Z. Order choice is fixed ex ante; prediction, abstention and continuation may still depend on observed tokens. Static

coupling preserves that distinction. Finite history size can grow exponentially with horizon. The LP is exact as a finite representation on its specified support, while real-data optimization is numerical.

Algorithm 3. Constrained finite-history realization LP with a strong ex ante-order control
Require: Hash-bound TRAIN-only ET54 argmax/max-confidence bank for mask0..7; TUNE labels and person groups for fitting only. Paid-history current-mask tokens, cuts .45/.60/.80, max3 purchases, price .05, wrong loss1, abstention .4. Equal-person mean coverage .50 and each fitting person .25; same rights for full adaptive and six-order static.
Ensure: A frozen common actor for each class, its complete supported-history table and ex ante order mixture where applicable. TUNE/CV/CAL person-level C,E,K,J, fallback/correction/harm/cost and sparse LP/process certificates; retained no_gain or failure.

1 Validate source hashes and documented numeric schemas. Read only allowed fields locally; never import the producer actor or model pickle.
2 For each fixed participant-CV fold, make the support graph and coefficients from fitting4 people only. Enumerate complete paid token histories, keeping prior actions and past tokens.
3 Create nonnegative root and history-action realization weights. Give every observation child the same parent-action upper bound, with leftover mass assigned to abstention.
4 For static add six ex ante order masses, root-axis coupling and first-history next-purchase caps. Keep observation-dependent prediction, stopping and abstention.
5 Minimize macro J with sparse LPs, then K, then maximize C on preceding optimum faces. Reconstruct conditional actions by child/parent ratios and verify person coefficients against the actor.
6 Evaluate each CV actor on its heldout4 using the fixed unsupported-history fallback. Select adaptive only if both OOF coverage gates pass and its unrounded deltaJ<-1e-9; otherwise static.
7 Fit both final classes on all TUNE8; Y-first is a fixed paired diagnostic only. Lock support, all weights, ties, files and the CV-selected class before reading CAL.
8 On CAL only evaluate the locked actors, literal controls and separately privileged label/future-aware oracle. Run fixed-seed cached controller replay without feeding truth/identity/unbought outputs to actor.
9 Preserve all denominators and resources. A useful CAL gain requires both original strict coverage floors and adaptive deltaJ<-1e-9; process/solver/resource failure is not no_gain.
10 Freeze reporting artifacts for a different actual reviewer. Keep exposure, finite-class, physical-cost and novelty limits; activate no skill or global setting.

$$J = 0.05K + E + 0.4(1 - C)$$

E is unconditional wrong-prediction mass, C prediction mass and K expected paid axes; purchases remain charged on final abstention.

$$a_0 + f_0 + \sum_A \theta_A = 1$$

Before any observation choose free abstention, fixed free-prior prediction or first axis A.

$$p_{A,s} + \sum_{B \neq A} u_{A,s,B} \leq \theta_A$$

Different observations s each have the same parent upper bound; they are not summed as policy-flow children.

$$v_{A,s,B,t} + w_{A,s,B,t} \leq u_{A,s,B}$$

The second paid history may predict, abstain or buy the sole remaining axis.

$$z_{A,s,B,t,r} \leq w_{A,s,B,t}$$

The full paid history terminates by current-mask prediction or abstention.

$$\theta_A = \sum_{o \in O_A} \lambda_o, \quad u_{A,s,B} \leq \lambda_{A,B,C}$$

Static order masses are chosen before observations. Observation-dependent stopping remains allowed.

$$\overline{C} = \left(\frac{1}{G}\right) \sum_{g=1}^{G} C_g \geq 0.50, \quad C_g \geq 0.25$$

The same fitting coverage conditions apply to both classes; CAL gate assesses them without refitting.

12. MHEALTH: conditional gain and stronger-control failure

The original empirical-risk adaptive and myopic policies have identical exposed DEV J=1364/4515 (about .302104). The selected fixed order attains 134663/457520 (about .294332) at the same full prediction coverage. The stronger historical committed token-2 static policy attains 28743/114380 (about .251294). Adaptive acquisition corrects 12 errors and introduces 5 harms relative to the selected order, yet buys 219 additional quotes. Against the stronger committed policy it corrects 5 and harms 32. Reduced errors against one control therefore do not establish the declared cost-sensitive gain.

Calibrated adaptive, myopic and selected fixed_32 tie exactly at 10141/39216 (about .258593) with identical paths. They abstain on 83 of 524 windows. Relative to empirical adaptive acquisition, 28 errors are corrected and 14 harms introduced on jointly predicted windows; the newly abstained cases include 42 prior errors and 41 prior correct predictions. Abstention is not counted as a prediction correction. The calibrated forced-subset reference is worse at 115999/343140 (about .338052), so comparing only to it would hide the strong historical control.

A coherent fixed-row weighting correction verifies 88 P/Q loss identities but changes none of 22 action tables. C-informed arbitration over a frozen finite menu yields paired gain zero. The complete supported-grammar learner then covers 4,836 adaptive forests and 1,064 order-tagged fixed forests. Its robust fixed reference has training endpoint vector (3/14,45/154). A mixture with weights 421/1471 and 1050/1471 certifies Γ =2091/205940 at common dual θ=162/1471. This positive statement is exact for the two empirical T-group laws.

On the same exposed DEV people, that paired mixture has J=316099513/1009517880 (about .313119), improves its original robust reference in the mean and still loses to T-mean, the selected historical order and the strongest committed control. Person 7 worsens; person 8 improves. All mixture totals and transitions are expectations of an unexecuted independent coin, not new randomized episodes.

The objective-matched complete shared-α receding-one-step family attains exactly the adaptive minimax value 123/440 and paired gain 2091/205940. Both primary full-planning increments are zero. A later known-θ frozen-row analysis has $V_M(\theta) - V_A(\theta) = \max\left(0, \frac{615\theta - 538}{6160}\right)$, reaching 1/80 at θ=1. In that region full fixed-order planning also attains the adaptive value. This is a conditional limitation of the specified M family, not adaptive query-choice superiority over F or a new DEV result.

Restoring the stronger historical static comparator and trying reference-first reserve routes does not produce a useful exposed-development acquisition gain. The exact reference-relative certificate, the strong-static deficit, the primary myopic ties and the post hoc known-law gap coexist because they answer different questions. Supplement S6 retains every policy table, person result, paired transition, objective decomposition, route/control identity and failure boundary.

Table 2: MHEALTH exposed DEV loss under the common error/abstention/quote ledger. These are dependent windows from the same two people.

Policy / selection	Exact J	Display	Interpretation
Historical committed token 2	28743/114380	.251294	Strongest retained historical static control
Forced two-additional global order	182801/686280	.266365	Distinct forced-purchase definition
Selected fixed 2 / T-mean F	134663/457520	.294332	Optional continuation
T-mean A = M	1364/4515	.302104	No full-planning increment
Paired A = M	316099513/ 1009517880	.313119	Positive T-hull certificate; DEV person harm
Robust A = M	145279/457520	.317536	Absolute minimax selection
Original robust F reference	115999/343140	.338052	Reference identity fixed before paired solve

13. WISDM: predictor gains, paid selection and fusion

Saved-model reconstruction verifies identical WISDM windows for the original 18-feature and fixed 54-feature full-mask classifiers. CAL equal-person accuracy increases from 25.08% to 33.39% (8.31 percentage points), with all eight people improving. The pooled comparison has 680 corrections and 199 harms, a different 7.47-point accuracy gain. TUNE mean accuracy rises from 28.07% to 30.76%, but two people regress. TRAIN is resubstitution. All 41 person outcomes and 18 activity-code error counts remain in Supplement S7.

The proper-mask family improves mean CAL accuracy at every nonempty paid mask relative to its original representation, while five of 56 CAL person-mask cells regress. Equal forest hyperparameters leave different effective capacity and extraction cost. Full-mask CAL J improves from .899199 to .816102 when the fixed three-axis cost .15 is added, but both are above all-abstain .4. In this full-purchase always-predict regime only, accuracy must exceed 75% to beat .4. This is a prediction improvement without an acquisition success.

The participant-excluded raw/raw-confidence acquisition study has no eligible learned family at useful coverage. Every unconstrained family selects all-abstain. The highest post hoc CAL coverage among frozen learned candidates is only 12.60% for raw features and 14.64% with confidence; their fixed purchases would require more than 37.5% and 25% coverage even with zero covered errors to beat .4. These necessary conditions diagnose those candidates, not every possible adaptive policy.

The HGB intervention completes seven TRAIN fits and 21 probability passes. Every nonempty mask has positive TUNE mean accuracy change, but all seven CAL mean changes regress relative to the shared 54-feature ET family. Its useful selected option has CAL coverage 49.78925361520835%, below 50% before rounding, and J=.541547058872187. No TUNE-selected option simultaneously retains useful coverage and J<.4 on CAL. Larger label-informed correctness unions remain diagnostic rather than achieved selectors.

The first paid-submask multinomial fusion attempt starts one of 55 planned calls and fails its fixed convergence contract at 18 inputs, not at the unattempted full 126-input design. Zero calls complete and 54 are unattempted. No efficacy or no_gain outcome follows from that execution. A numerical cause was not established. The separate class-bias/simplex successor completes all 55 prescribed calls without retraining the base classifiers.

The successor's frozen useful mixture choice is Y-only at threshold .45 in every budget and is exactly the calibrated own-mask choice. On CAL both give J=.538803120435, mean coverage .507772045856 and minimum-person coverage .361904761905. Thus the primary mixture-versus-calibrated-own contrast is no_gain. It improves on one feasible calibrated log-pool choice at budgets 2 and 3, while other comparisons are ties or reference-infeasible. That narrower positive comparison does not turn the primary tie into mixture superiority. None of the useful choices beats free all-abstain.

Calibration and probability fusion may change argmax and raw expert-union membership. Paired accepted-error decompositions, common-acceptance comparisons, raw versus calibrated unions, person/class/fold changes and the descriptive eight-person bootstrap are preserved in Supplement S9. The bootstrap describes these reused people; it does not repair repeated development selection, multiplicity or small sample size.

14. Observable acquisition and nominal coverage optimization

The frozen confidence-.45 observable-history experiment isolates acquisition after terminal calibration. Full depth retains all paid-submask tokens and coherent person-weighted conditional masses. Its strongest controls include the same calibrated fixedY45 terminal, all Y-including subsets, all six global orders and equal stopping/support rights. The empirical-stopping diagnostic changes terminal rights and is reported separately. Its outcome cannot establish the primary acquisition effect.

The Y-first full-horizon candidate has CAL J=.466022, C=.310066, minimum-person C=.211111 and K=1.426129. Full-horizon and matched myopic aggregate results coincide. Its loss is lower than fixedY45 by .072781, but it fails both useful-coverage floors. The strong TUNE-selected unconstrained Y-first order has still lower J=.464206; candidate-minus-order is +.001816 and both are infeasible. No useful-feasible Y-first order exists in the TUNE selection menu, so that primary comparator remains unavailable rather than being replaced from CAL.

The principal visible failure is inadequate prediction coverage and paid cost on abstained cases. Observable branching can correct a baseline error or damage a correct prediction; both paired directions remain in the source tables. The calibrated fixedY control already achieves useful CAL mean/person coverage. A sequential policy that changes terminal acceptance, acquisition depth or anchoring must be compared within those same rights rather than against a weaker all-abstain or uniform-order control. Complete control and path/cost ledgers are in Supplement S10.

Against fixedY45, the full-horizon candidate corrects 118 jointly accepted errors and introduces 45 harms. It newly predicts on 69 windows (53 correct,16 wrong) and removes 1,371 predictions (287 correct,1,084 wrong). It spends 6,007 pooled queries on cases that ultimately abstain. These transitions explain why a smaller accepted-error mass is not a fixed-coverage accuracy improvement. The .05 δ K, δ E and -.4 δ C terms are +.021306, -.173170 and +.079082, respectively.

The nominal finite-library LP asks whether randomization among these frozen policies restores useful coverage fairly. It imposes TUNE mean C>=.50 and every TUNE person C>=.25 on both the eleven-ID static and fourteen-ID enriched libraries. It permits individually infeasible components because coverage is linear after convexification. The same loss and tie rules apply to both; enriched contains static, so its fitted optimum cannot be worse in exact arithmetic.

Both nominal mixtures reach the fitted mean boundary but fail it on CAL. CAL C is .446729 for static and .446425 for enriched. The unrounded enriched-minus-static J is approximately +.000509612. Every

primary CAL person still exceeds the .25 floor, so the mean condition is the binding transfer failure. A small fitted objective advantage and the superset relationship do not preserve evaluation ranking.

The equal-person ratio of total accepted error to total coverage differs from the mean of the eight person-specific ratios and from pooled-window selective error. Expected mixture purchases and fractional transition counts refer to the independent coin. Aggregate policy counts alone do not identify all cross-policy correction/harm intersections; the nominal LP audit does not invent them. Its diagnostic oracle expands information and is kept distinct from deployable common mixing. Supplement S11 gives the convexification proof, independent rational check, full person expectations, branch costs, aliases and nonidentifiability limits.

15. Known-person robustness does not transport useful coverage

The final study keeps the same frozen static and enriched libraries and common ex ante policy coin. It replaces uniform TUNE weighting with the eight vertices $q_v = (1-\rho)u + \rho e_v$, where u is uniform over the eight known TUNE people. At each vertex it constrains mean coverage at least .50, retains each original person floor .25, and minimizes worst-vertex J. The same vertex combines E,C,K into coherent loss; it never independently maximizes their components. The primary ρ is 1/4, with fixed sensitivities 0,1/8,1/2,1.

At ρ=1/4, the static optimum is .905741429286 fixedY45 plus .094258570714 order124/Yfirst. The enriched optimum is .906771494229 fixedY45 plus .093228505771 depth1/Yfirst. Both fitted minimum-vertex coverages equal .50 within errors below 6e-15. TUNE worst J is .590699703272 and .590383742672, respectively. This weak fitted enriched advantage follows the library containment; it is not a transport theorem.

The frozen CAL mean coverages are .489185592512 and .489340248133, short of .50 by .010814407488 and .010659751867. Every CAL person exceeds .25. CAL equal-person J is .531771719061 for static and .532017868971 for enriched, so δ J is +.000246149910. The enriched policy has slightly higher coverage and higher loss. Both useful-coverage comparisons fail, and the scientific outcome is no_gain with generalization still needing evidence.

The deterministic fixedY45 control is genuinely feasible under the same robust TUNE problem: fitted worst-vertex C=.522762038538 and minimum-person C=.350187265918. On CAL it passes the original mean/person gate with C=.507772045856, minimum C=.361904761905 and J=.538803120435. The randomized optima have lower CAL J but fail coverage; that lower loss is not a useful-domain gain. The original gate is distinct from a newly imagined adversarial-CAL gate, which fixedY45 need not pass.

At ρ=1/8 both means remain below .50. At ρ=1/2 neither library is feasible; the minimum common coverage relaxation and matching dual bound are .034762885669. At ρ=1 they are .149812734082. Those are positive finite-library infeasibility certificates, not timeouts. No CAL outcome is assigned to a nonexistent selected mixture. Expanding ρ shrinks the feasible set; exact feasible worst loss cannot decrease.

The conditional coverage result is elementary but narrow. If a target's full policy-coverage response vector equals a TUNE-person reweighting in the declared uncertainty set, any feasible common coin transports its .50 expected coverage. The primary static CAL mean response violates that premise, witnessed by the selected linear functional. A separate coordinate witness has CAL person 7 static7 coverage 259/1602 below the TUNE minimum 266/1602, excluding that finite response vector even from the full observed TUNE hull. This is not population OOD detection, a causal diagnosis or a statement that every CAL vector lies outside the hull.

TUNE uses out-of-fold participant-pair calibration outputs, whereas CAL uses the all-TUNE refit. The exposed CAL group also differs from TUNE. Aggregate counts cannot isolate how much of the failure is due to person shift, refit change, prediction errors or catalogue restrictions. No nested untouched outer evaluation or new-person inclusion probability is supplied. Supplement S12 retains all primary person rates, all radii, exact control IDs, locking and poison-selector checks, certificates and process accounting.

$$\mathcal{Q}_\rho = \mathrm{conv}\{(1-\rho)u + \rho e_v : v = 1, ..., 8\}, \quad \min_{q \in \mathcal{Q}_\rho} \sum_g q_g C_g(w) \geq \left(\frac{1}{2}\right)$$

Known-person reweighting set. ρ interpolates among eight observed response vectors, not among unknown populations.

$$\min_{w,z} z: \quad q_v^T J w \leq z, \quad q_v^T C w \geq \frac{1}{2}, \quad C_g w \geq \frac{1}{4}, \quad w \in \Delta(\Pi)$$

One coherent robust epigraph program per library, with all vertex and person constraints and the shared independent coin.

Table 3: Locked primary CAL results at ρ=1/4. Values are expectations; original feasibility uses unrounded mean/person rates.

Policy	Mean J	Mean C	Min person C	Original floors
Randomized static11	.531771719061	.489185592512	.348738	fail mean
Randomized enriched14	.532017868971	.489340248133	.347846	fail mean
TUNE-chosen fixedY45	.538803120435	.507772045856	.361904761905	pass
Uniform six orders	.465276	.302284	.188095	fail both

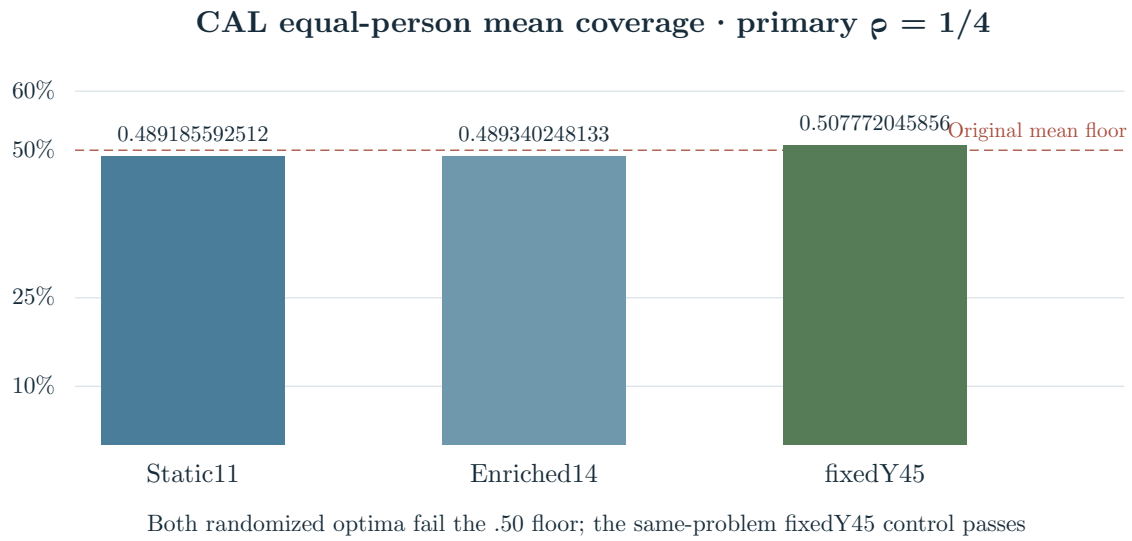


Figure 4: Primary ρ=1/4 coverage on exposed CAL. Both optimally fitted randomized libraries miss the original .50 mean floor, while the same-problem feasible fixedY45 control passes. Lower loss under a failed coverage gate does not establish a useful-domain win. The plot is drawn from frozen aggregate values.

16. Conditional control: useful coverage passes, but development loss regresses

TUNE adaptive J=0.465403574 improves over static 0.474064787 while both mean coverages are approximately.50. On participant OOF, however, adaptive J=0.648135368 is worse than static 0.614714591 by 0.033420777696. Both OOF coverage gates pass. The locked practical selector therefore chooses full_static.

A generic comparison helper left a reporting-label error in the immutable raw TUNE object: it calls the fitted optimum difference a limited exposed-development gain and uses CAL-named coverage keys. REPORTING_SCOPE_CORRECTION.json binds the original result hash and corrects this interpretation to fit-only optimum difference. TUNE improvement is not heldout useful gain. Raw numbers/code/logs are preserved and were not rerun or edited.

On reused CAL the adaptive/static mean coverages are 0.579668276/0.569936044 and minimum-person coverages 0.481599895/0.459028767. Both original strict floors pass. Adaptive error mass is 0.328937230 versus 0.311608279, purchases 1.930761711 versus 1.822020128. The unrounded loss difference is +0.018873137736314383, so this result is no_gain with regression. The CV-selected actor is exactly the final static actor; its improvement relative to itself is zero.

Coverage is constrained by the same lower floors, not forced equal in realization. Adaptive predicts on more CAL mass and also makes more wrong predictions and purchases. The comparison is a common constrained-objective contrast, not an equal-realized-coverage risk curve. The Y-first paired diagnostic gives the same adaptive policy and effectively the same static result up to numerical error; it was not used to choose a winner.

The CAL ledger gives deltaK=0.1087415836059491, deltaE=0.0173289514759441 and deltaC=0.0097322322998179. Thus the extra purchase term is approximately +0.005437079180, error +0.017328951476 and abstention reduction -0.003892892920, summing to +0.018873137736. This is an accounting decomposition, not a causal identification of why the controller transferred poorly.

Unsupported-history prediction mass is 0.10230956689761754 for adaptive and 0.10610456189286122 for static. The frozen current-mask fallback may add coverage and error, but these two aggregate masses cannot by themselves isolate the cause of regression. A high-coverage controller can still have high unconditional and selective error.

The raw generic TUNE/CV comparison labels are interpreted through REPORTING_SCOPE_CORRECTION.json: TUNE δ is fit-only; CV is participant OOF development; neither is CAL efficacy. The original numeric artifact and scientific code are unchanged.

Table 4: Same TRAIN-only bank, analytic policy expectations. All8people retained; E is unconditional.

Split	Class	J	C	E	K	Min C
TUNE	adaptive	0.465403574	0.500000000	0.171040798	1.887255513	0.400516550
TUNE	static	0.474064787	0.500000000	0.184219001	1.796915730	0.407557529
CAL	adaptive	0.593608005	0.579668276	0.328937230	1.930761711	0.481599895
CAL	static	0.574734868	0.569936044	0.311608279	1.822020128	0.459028767

Table 5: Prespecified twofold participant OOF on TUNE; different fold actors, not a new cohort.

Class	J	C	E	K	Min C
adaptive	0.648135368	0.614976206	0.402471158	1.833093864	0.533333333
static	0.614714591	0.577029946	0.359699664	1.716538096	0.470005136

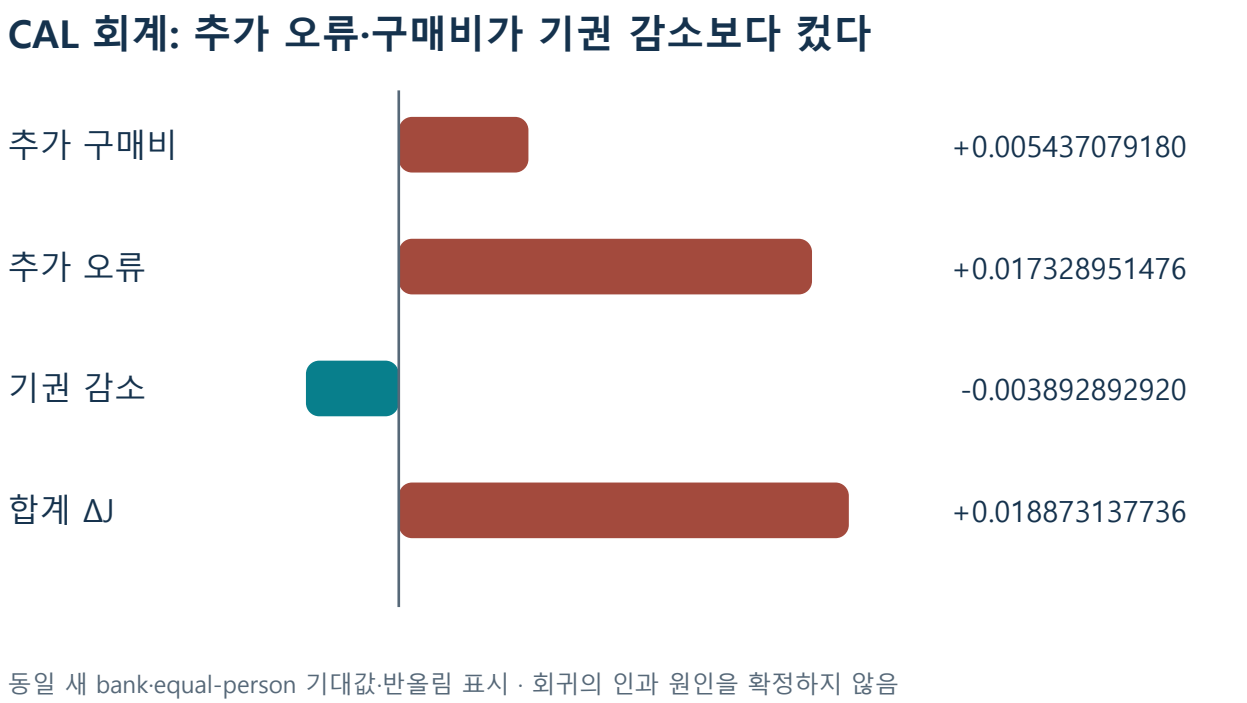


Figure 5: CAL adaptive-minus-static loss ledger. Extra errors and logical purchase cost outweigh the reduction in abstention loss. Terms are rounded for display; this decomposition is descriptive, not causal.

17. A separate posthoc diagnosis: small class potential is outweighed by transfer excess

After the known regression, a separately registered six-LP diagnosis fits the same observable adaptive/static classes to CAL labels. It does not modify the frozen TUNE actors or the prespecified CV choice. The CAL-fit double objectives are .45644611777376054 and .459360117059504; their reported contrast is -.00291399928574346. CAL labels are now fitting inputs, so these values are not independent test performance.

The unchanged TUNE actors embed into CAL support with all original C/E/K/J expectations preserved; unsupported-history fallback is retained. Their excess losses relative to the fitted class optima are .13716188770309165 and .1153747506810338. The excess-difference +.02178713702205786 outweighs the small negative class contrast, reconstructing the original positive CAL δ. This is a same-finite-law decomposition, not a causal identification of person shift or proof of general learnability.

The first independent diagnosis review found exact-bound reporting failures. A separate completed arithmetic reconciliation, using no LP and no raw cache reads, identifies an empty integer sum converting the free-prior metric and upper witness to float. Both original lower rationals match, but both reported upper strings differ from their claimed exact witness values. The corrected rational bounds retain a negative class contrast,

approximately [-.002913999285755284,-.002913999285743204]. This sign is a privileged finite-law potential statement.

The raw CAL-fit static mean coverage is about8.5e-15 below .50. A proof-only prior mixture of approximately1.695077125254431e-14 supplies a strictly feasible exact upper witness; it is not a change to the stored actor or reported performance. The original static decimal upper was below that witness and is not certified by it; this does not establish that it lies below the unknown true optimum. Use the linked corrected rationals. The original failed review, read-budget failure and immutable source artifacts remain visible in Supplement S16.

$$t_A - t_F = (t_A - J_A^\star) - (t_F - J_F^\star) + (J_A^\star - J_F^\star)$$

A formal excess decomposition for the same empirical objective. Displayed fitted-double scalars and exact-optimum bound intervals have separate scopes.

Table 6: Same exposed CAL law,8people/6437dependent windows. Raw fitted doubles and frozen-actor expectations; posthoc oracle access is explicit.

Quantity	Value	Information/interpretation
CAL-fit adaptive J	.45644611777376054	CAL-label-fitted same observable class
CAL-fit static J	.459360117059504	CAL-label-fitted six-order class
Reported class contrast	-0.0029139992857434602	Privileged posthoc potential, not deployed gain
Excess-difference	0.021787137022057861	Frozen TUNE actors minus class optima
Original parent CAL δJ	+.018873137736314383	Unchanged held-development regression

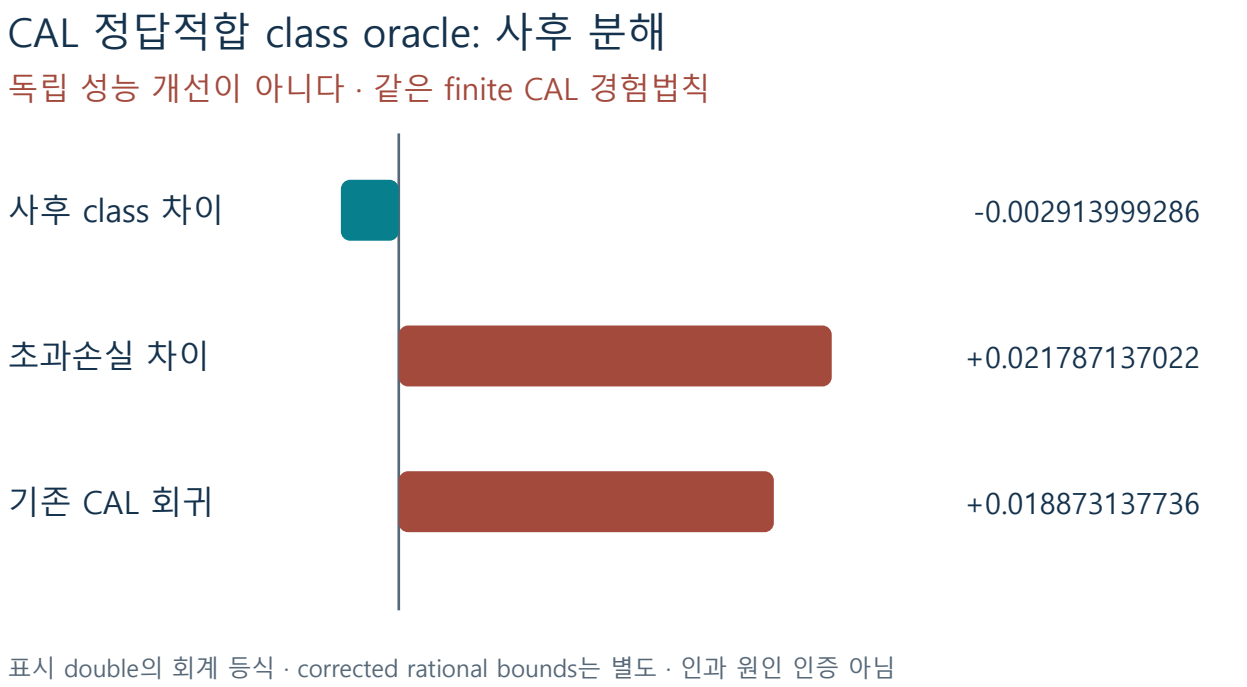


Figure 6: CAL-label-fitted same-observable-class oracle diagnosis. A small negative class contrast is outweighed by a larger positive frozen-policy excess contrast. The result is posthoc potential, not a deployment improvement.

18. Computation, preserved failures and reproducibility

Complete enumeration is exponential in query depth and observation alphabet. The two-law solver's pair search, skyline work, exact rational bit cost and all-original-row witness check are separate operations. Its preserved caps include 2,000 policies per class, 64 skyline rows, 5,000 candidates per solve and 4,096-bit serialized fractions. A cap violation fails explicitly. General multiworld/group coverage problems use an LP and cannot inherit the two-row-support reduction.

Finite controlled constrained-world computation retains 9,579 syntactic root trees across ten cases and 43 LPs. This is exact small-model evidence, not a scalable learner. The empirical coverage/transport LP operates on 11 and 14 frozen policy IDs rather than all possible adaptive trees. Its certificate validates that finite catalogue and aggregate arithmetic, not the accuracy of the upstream predictors or a missing policy's performance.

Measured empirical resources are scope-specific. The matched acquisition child reports 1,008 risk fits, 2,128 prediction passes and 11,415,264 row-model predictions, with observed exit 0 and CPU 274.578 seconds. HGB reports seven fits/21 probability passes and terminal CPU 189.046875 seconds. The completed bias/mixture child reports 55 calls and CPU 153.453125 seconds. These one-attempt measurements have different inputs and tasks; they are not a matched speed benchmark or physical sensing cost.

The failure ledger retains the original MHEALTH terminal-mask/domain failure, the WISDM four-stream clock/alignment failure, representation serialization failure, the failed predecessor verification, missing historical launcher/terminal observations, first-fit multinomial nonconvergence, nominal and robust coverage failures, and larger-radius infeasibility. A successful later durable derivative does not retroactively make an earlier failed process successful. Source and process custody support specific claims rather than general scientific certification.

The conditional child completes 26 LPs: eight three-stage constrained controller fits (24) and two separately privileged split mask-oracle problems. The 24 fit stages retain full coefficient/primal/dual certificates; the two oracle summaries do not retain full x/dual vectors and require their separately reported independent arithmetic scope. These are different evidence depths, not 26 full-dual independent verifications. The measured child reports 79.469s wall,71.015625s CPU and139259904bytes peak. No model fit or inference occurs; native authoring remains unmetered.

This editorial build performs zero scientific model/API calls and reruns zero experiments. Historical experimental API-call counts of zero concern those bounded computational children. Native model-assisted authoring usage was not metered and must not be called zero cost. Source retrieval, authoring, coordinator work, rendering and whole-host load are outside most experimental counters. Watchdogs and subprocess timeouts are not OS resource or security isolation.

Editable DOCUMENT, Typst, diagrams, source/claim mapping and build/QC records accompany the two PDFs. Each selected input is hash-bound and preserved unchanged; local source scripts are treated as data and are not run. Supplement S13 states the full resource, custody, independent-check and reproduction boundaries. Reproducibility here includes a rebuildable editorial artifact and inspectable frozen evidence, without claiming independent refits, private-data release, external benchmark reproduction or actual submission.

The primary review decision is needs_evidence: its24fit-LP/40metric/79,982history and two finite-oracle numerical checks pass, while its reading-resource contract fails. The first diagnosis review retains its failed exact-bound checks. The separate arithmetic reconciliation is accepted only within its bounded no-LP/no-raw-cache scope. Neither a numerical check nor that scoped acceptance certifies all resources, population validity or novelty.

19. Limitations and interpretation

The strongest positive results are conditional finite-model statements. They depend on declared worlds, shared observation and action rights, correct complete enumeration and exact or checked numerical certificates. They do not estimate true-world inclusion. The two-law MHEALTH empirical certificate concerns two observed training-group laws and one fixed reference; it neither dominates every stronger exposed-development static policy nor establishes a general full-planning benefit.

The empirical record remains development. MHEALTH uses two exposed DEV people with dependent windows. WISDM TUNE/CAL have informed repeated method design, and global FINAL independence is unresolved. TRAIN resubstitution, TUNE OOF estimates, all-TUNE refits, CAL selection/reporting and ancestry-specific prediction banks have different meanings. No repeated-window count is promoted to an independent sample size, and no unexecuted final result is inferred.

Logical axes, tokens and quotes are accounting units. The experiments measure neither independent sensor activation, energy, latency, communication, reliability nor online segmentation. Cached counterfactual values permit retrospective evaluation under the declared bank but do not demonstrate physical provider behavior or missingness identification. The frozen support fallbacks can limit performance, and changing them would define a new study.

Prediction-family comparisons change representation or learner capacity and compute. Terminal calibration can change predicted labels. Empirical-stopping and unanchored diagnostics change rights. Each is reported within its scope; none is used to explain all acquisition failures causally. Aggregate policy counts do not identify every joint correction/harm event or separate person shift from refit shift.

Known-person robustness raises fitted coverage pressure but cannot manufacture an inclusive target response set. The latest primary failure and finite hull witness invalidate particular observed-set premises; they do not establish a general population OOD classifier or an impossibility theorem for every future policy. Higher-

ρ infeasibility is restricted to the frozen finite libraries. A wider conditional TV certificate is not measured robust prediction gain.

A confirmatory acquisition claim would require a separately governed untouched participant cohort, a prespecified full pipeline and useful-coverage/cost target, justified world-set or estimator uncertainty, stronger matched-capacity and learned-method controls, and independent review of realized harms and failures. Those experiments are not performed or promised as completed by this manuscript. The present contribution is a complete, inspectable boundary between exact conditional theory and the available negative development evidence.

The richer class has a genuine fitted advantage under the new finite contract, but participant CV and CAL reverse its ranking. The old transport coverage failure and the new coverage-feasible regression are distinct negative results on different predictor/terminal views. Label-aware hindsight and any CAL-fitted observable-policy diagnosis add outcome access and remain posthoc. They cannot be reported as a deployed independent gain or used to retune the locked selector.

Corrected finite-CAL class bounds distinguish absence of observable-class potential from loss incurred by the fitted transferred policies, but do not identify a unique causal mechanism or establish sample-efficient learnability. Outcome access, fixed representation, eight historically exposed people and logical costs delimit the diagnosis. Raw floating actors, normalized proof witnesses and reported decimal summaries are distinct objects.

20. Conclusion

ProbeFlow provides a common language for paid information, strong comparators, same-world gain and useful coverage. Exact finite routed models exhibit real conditional adaptive value and decisive no-gain regimes. Empirical-law certificates survive their mathematical checks while failing to establish the desired independent application benefit: strong MHEALTH static controls win, the complete declared myopic family matches both primary adaptive objectives, and WISDM predictor improvements do not translate into a successful matched acquisition comparison.

Common randomization and robust known-person constraints change the feasible problem and its fitted optimum, but the final frozen evaluation still misses useful mean coverage and favours the optimized static control in loss. The feasible fixedY45 alternative and positive larger-radius infeasibility certificates make the failure specific and falsifiable. Complete proofs, participant outcomes, counterexamples and process failures support these conclusions within their stated evidence boundaries.

Optimizing the richer supported conditional class closes one catalogue limitation without delivering the desired practical win. Both new CAL coverage gates pass, yet extra errors and purchases dominate its additional prediction coverage; the prespecified development selector chooses the matching static class. The next falsifiable step requires independently governed people excluded from all design and selection, with the full information, budget, terminal and useful-coverage contract locked first.

A small negative posthoc class contrast survives the corrected empirical-law bounds, while the larger transfer-excess contrast reconstructs the actual regression. This narrows one finite-law explanation without turning a CAL-label-fitted oracle into a practical win. Conditional finite theory and predictor changes remain separate from acquisition-strategy and independent physical efficacy.

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